

<h1 id="3D-Array">3D Array</h1>

In [1]: `import numpy as np`

```
arr = np.array([
    [
        [10, 11, 12, 13, 14],
        [15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24],
        [25, 26, 27, 28, 29]
    ],
    [
        [30, 31, 32, 33, 34],
        [35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44],
        [45, 46, 47, 48, 49]
    ],
    [
        [50, 51, 52, 53, 54],
        [55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64],
        [65, 66, 67, 68, 69]
    ]
])

print(arr.shape)
```

(3, 4, 5)

In [2]: `import numpy as np`
`print(np.__version__)`

2.2.6

In [3]: *# Assignment: 3D Array Indexing & Slicing*

Section A - Access Individual Elements

```
# Write a statement to extract:
```

```
10  
24  
37  
43  
54  
66  
58  
49
```

Out[3]: 49

```
In [4]: print(arr[0, 0, 0]) #Access Element (0, 0, 0):",  
print(arr[0, 2, 4]) #"Access Element (0, 2, 4):",  
print(arr[1, 1, 2]) #"Access Element (1, 1, 2):",  
print(arr[1, 2, 3]) #"Access Element (1, 2, 3):",  
print(arr[2, 0, 4]) #"Access Element (2, 0, 4):",  
print(arr[2, 3, 1]) #"Access Element (2, 3, 1):",  
print(arr[2, 1, 3]) #"Access Element (2, 1, 3):",  
print(arr[1, 3, 4]) #"Access Element (1, 3, 4):",
```

```
10  
24  
37  
43  
54  
66  
58  
49
```

```
In [5]: # Section B - Access Complete Rows
```

```
# Write a statement to extract:
```

```
1.  
[15, 16, 17, 18, 19]  
  
2.  
[40, 41, 42, 43, 44]
```

```
3.  
[65, 66, 67, 68, 69]
```

Out[5]: [65, 66, 67, 68, 69]

```
In [6]: print(arr[0,1])  
        print(arr[1,2])  
        print(arr[2,3])
```

```
[15 16 17 18 19]  
[40 41 42 43 44]  
[65 66 67 68 69]
```

```
In [8]: # Section C - Access Complete 2D Matrices
```

```
# Extract:
```

```
# The first matrix
```

```
# The second matrix
```

```
# The third matrix
```

```
In [9]: print("The first matrix\n\n",arr[0])  
        print("The second matrix\n\n",arr[1])  
        print("The third matrix\n\n",arr[2])
```

The first matrix

```
[[10 11 12 13 14]
 [15 16 17 18 19]
 [20 21 22 23 24]
 [25 26 27 28 29]]
```

The second matrix

```
[[30 31 32 33 34]
 [35 36 37 38 39]
 [40 41 42 43 44]
 [45 46 47 48 49]]
```

The third matrix

```
[[50 51 52 53 54]
 [55 56 57 58 59]
 [60 61 62 63 64]
 [65 66 67 68 69]]
```

```
In [10]: # Section D - Column Extraction
```

```
1
```

```
[10, 15, 20, 25]
```

```
2
```

```
[32, 37, 42, 47]
```

```
3
```

```
[54, 59, 64, 69]
```

```
Out[10]: [54, 59, 64, 69]
```

```
In [11]: print(arr[0,:,0])
```

```
print(arr[1,:,2])
```

```
print(arr[2,:,4])
```

```
[10 15 20 25]
```

```
[32 37 42 47]
```

```
[54 59 64 69]
```

```
In [ ]: # # Section E - Slicing

# Extract:

# The first two matrices.
# The last two matrices.
# The first two rows from every matrix.
# The last two rows from every matrix.
# The first three columns from every matrix.
# The last two columns from every matrix.
```

```
In [12]: print(arr[0,0:2,:])
print(arr[0,2,:])
print(arr[1,0:2,:])
print(arr[1,2,:])
print(arr[2,0:2,:])
print(arr[2,2,:])
print(arr[0,0:2,0:2])
print(arr[1,0:2,0:2])
print(arr[2,0:2,0:2])
print(arr[0,:,0:3])
print(arr[1,:,0:3])
print(arr[2,:,0:3])
print(arr[0,:,3:])
print(arr[1,:,3:])
print(arr[2,:,3:])
```

```
[[10 11 12 13 14]
 [15 16 17 18 19]]
[[20 21 22 23 24]
 [25 26 27 28 29]]
[[30 31 32 33 34]
 [35 36 37 38 39]]
[[40 41 42 43 44]
 [45 46 47 48 49]]
[[50 51 52 53 54]
 [55 56 57 58 59]]
[[60 61 62 63 64]
 [65 66 67 68 69]]
[[10 11]
 [15 16]]
[[30 31]
 [35 36]]
[[50 51]
 [55 56]]
[[10 11 12]
 [15 16 17]
 [20 21 22]
 [25 26 27]]
[[30 31 32]
 [35 36 37]
 [40 41 42]
 [45 46 47]]
[[50 51 52]
 [55 56 57]
 [60 61 62]
 [65 66 67]]
[[13 14]
 [18 19]
 [23 24]
 [28 29]]
[[33 34]
 [38 39]
 [43 44]
 [48 49]]
[[53 54]
 [58 59]]
```

```
[63 64]
[68 69]]
```

```
In [13]: # Section F - Advanced Slicing
```

```
[
    [10, 11, 12],
    [15, 16, 17]
]
```

```
Out[13]: [[10, 11, 12], [15, 16, 17]]
```

```
In [14]: print(arr[0,0:2,0:3])
```

```
[[10 11 12]
 [15 16 17]]
```

```
In [15]: [
    [42, 43, 44],
    [47, 48, 49]
]
```

```
Out[15]: [[42, 43, 44], [47, 48, 49]]
```

```
In [16]: print(arr[1,2:,2:])
```

```
[[42 43 44]
 [47 48 49]]
```

```
In [17]: [
    [55, 56],
    [60, 61],
    [65, 66]
]
```

```
Out[17]: [[55, 56], [60, 61], [65, 66]]
```

```
In [18]: print(arr[2,1:,0:2])
```

```
[[55 56]
 [60 61]
 [65 66]]
print(arr[2,1:3,0:1])
```

```
In [19]: [
          [13, 14],
          [18, 19]
        ]
```

```
Out[19]: [[13, 14], [18, 19]]
```

```
In [20]: print(arr[0,0:2,3:])
```

```
[[13 14]
 [18 19]]
```

```
In [21]: print(arr[:,::2,::2])
```

```
[[[10 11 12 13 14]
   [15 16 17 18 19]
   [20 21 22 23 24]
   [25 26 27 28 29]]
```

```
[[30 31 32 33 34]
 [35 36 37 38 39]
 [40 41 42 43 44]
 [45 46 47 48 49]]
```

```
[[50 51 52 53 54]
 [55 56 57 58 59]
 [60 61 62 63 64]
 [65 66 67 68 69]]]
```

```
In [23]: # Section G - Questions
```

```
# Write a single slicing statement to extract:
```

```
# Every alternate matrix.
```

```
# Every alternate row from all matrices.
```

```
# Every alternate column from all matrices.
```

```
# The first column from every matrix.
# The last column from every matrix.
# The middle row from every matrix.
# The middle column from every matrix.
# Reverse the order of matrices.
# Reverse the rows inside every matrix.
# Reverse the columns inside every matrix.
```

In []:

<h1 id="2D-ARRAY">2D ARRAY</h1>

In [24]: **import** numpy **as** np

```
arr = np.array([
    [10, 11, 12, 13, 14, 15],
    [20, 21, 22, 23, 24, 25],
    [30, 31, 32, 33, 34, 35],
    [40, 41, 42, 43, 44, 45],
    [50, 51, 52, 53, 54, 55],
    [60, 61, 62, 63, 64, 65]
])

print(arr.shape)
```

(6, 6)

In [26]: *## 2D Array Indexing & Slicing*

```
# Extract the following using a single indexing/slicing statement
# 1. Every alternate row

# Expected Output:

# [
#  [10,11,12,13,14,15],
#  [30,31,32,33,34,35],
#  [50,51,52,53,54,55]
# ]
```

```
In [27]: print(arr[0::2,0:::])
```

```
[[10 11 12 13 14 15]
 [30 31 32 33 34 35]
 [50 51 52 53 54 55]]
```

```
In [29]: # # 2. Every alternate column
```

```
# Expected Output:
```

```
# [
#  [10,12,14],
#  [20,22,24],
#  [30,32,34],
#  [40,42,44],
#  [50,52,54],
#  [60,62,64]
# ]
```

```
In [30]: print(arr[0::,0:5:2])
```

```
[[10 12 14]
 [20 22 24]
 [30 32 34]
 [40 42 44]
 [50 52 54]
 [60 62 64]]
```

```
In [32]: # 3. First column
```

```
# Expected Output:
```

```
# [10,20,30,40,50,60]
```

```
In [33]: print(arr[0:6:1,0:2:4])
```

```
[[10]
 [20]
 [30]
 [40]
 [50]
 [60]]
```

```
In [35]: # 4. Last column

# Expected Output:

# [15,25,35,45,55,65]
```

```
In [36]: print(arr[0:6:1,5:6:1])
```

```
[[15]
 [25]
 [35]
 [45]
 [55]
 [65]]
```

```
In [38]: # 5. Middle row

# Expected Output:

# [40,41,42,43,44,45]
```

```
In [39]: print(arr[3:4:3,:])
```

```
[[40 41 42 43 44 45]]
```

```
In [41]: # 6. Middle column

# Expected Output:

# [13,23,33,43,53,63]
```

```
In [42]: print(arr[:,3:4:3])
```

```
[[13]
 [23]
 [33]
 [43]
 [53]
 [63]]
```

```
In [43]: # 7. First three rows
```

```
# Expected Output:
```

```
# [
#  [10,11,12,13,14,15],
#  [20,21,22,23,24,25],
#  [30,31,32,33,34,35]
# ]
```

```
In [44]: print(arr[0:3,:,:])
```

```
[[10 11 12 13 14 15]
 [20 21 22 23 24 25]
 [30 31 32 33 34 35]]
```

```
In [45]: # 8. Last three rows
```

```
# Expected Output:
```

```
# [
#  [40,41,42,43,44,45],
#  [50,51,52,53,54,55],
#  [60,61,62,63,64,65]
# ]
```

```
In [46]: print(arr[3:6,:,:])
```

```
[[40 41 42 43 44 45]
 [50 51 52 53 54 55]
 [60 61 62 63 64 65]]
```

```
In [47]: # 9. First three columns
```

```
# Expected Output:
```

```
# [  
#  [10,11,12],  
#  [20,21,22],  
#  [30,31,32],  
#  [40,41,42],  
#  [50,51,52],  
#  [60,61,62]  
# ]
```

```
In [48]: print(arr[:,0:3:])
```

```
[[10 11 12]  
 [20 21 22]  
 [30 31 32]  
 [40 41 42]  
 [50 51 52]  
 [60 61 62]]
```

```
In [50]: # 10. Last three columns
```

```
# Expected Output:
```

```
# [  
#  [13,14,15],  
#  [23,24,25],  
#  [33,34,35],  
#  [43,44,45],  
#  [53,54,55],  
#  [63,64,65]  
# ]
```

```
In [51]: print(arr[:,3:6:])
```

```
[[13 14 15]
 [23 24 25]
 [33 34 35]
 [43 44 45]
 [53 54 55]
 [63 64 65]]
```

```
In [52]: # 11. Reverse all rows
```

```
# Expected Output:
```

```
# [
#  [60,61,62,63,64,65],
#  [50,51,52,53,54,55],
#  [40,41,42,43,44,45],
#  [30,31,32,33,34,35],
#  [20,21,22,23,24,25],
#  [10,11,12,13,14,15]
# ]
```

```
In [53]: print(arr[7::-1,:])
```

```
[[60 61 62 63 64 65]
 [50 51 52 53 54 55]
 [40 41 42 43 44 45]
 [30 31 32 33 34 35]
 [20 21 22 23 24 25]
 [10 11 12 13 14 15]]
```

```
In [54]: # 12. Reverse all columns
```

```
# Expected Output:
```

```
# [
#  [15,14,13,12,11,10],
#  [25,24,23,22,21,20],
#  [35,34,33,32,31,30],
#  [45,44,43,42,41,40],
#  [55,54,53,52,51,50],
```

```
# [65,64,63,62,61,60]
# ]
```

```
In [55]: print(arr[:,6::-1])
```

```
[[15 14 13 12 11 10]
 [25 24 23 22 21 20]
 [35 34 33 32 31 30]
 [45 44 43 42 41 40]
 [55 54 53 52 51 50]
 [65 64 63 62 61 60]]
```

```
In [56]: # 13. Reverse rows and columns simultaneously
```

```
# Expected Output:
```

```
# [
# [65,64,63,62,61,60],
# [55,54,53,52,51,50],
# [45,44,43,42,41,40],
# [35,34,33,32,31,30],
# [25,24,23,22,21,20],
# [15,14,13,12,11,10]
# ]
```

```
In [57]: print(arr[6::-1,6::-1])
```

```
[[65 64 63 62 61 60]
 [55 54 53 52 51 50]
 [45 44 43 42 41 40]
 [35 34 33 32 31 30]
 [25 24 23 22 21 20]
 [15 14 13 12 11 10]]
```

<h1 id="Write-a-single-indexing/slicing-statement-to-extract:">Write a single indexing/slicing statement to extract: </h1>

```
In [58]: [
          [22,23,24],
          [32,33,34],
```

```
[42,43,44]  
]
```

Out[58]: [[22, 23, 24], [32, 33, 34], [42, 43, 44]]

In [59]: print(arr[:,::])

```
[[10 11 12 13 14 15]  
 [20 21 22 23 24 25]  
 [30 31 32 33 34 35]  
 [40 41 42 43 44 45]  
 [50 51 52 53 54 55]  
 [60 61 62 63 64 65]]
```

In [60]: print(arr[1:4:1,2:5:1])

```
[[22 23 24]  
 [32 33 34]  
 [42 43 44]]
```

In [61]: [
 [11,13,15],
 [31,33,35],
 [51,53,55]
]

Out[61]: [[11, 13, 15], [31, 33, 35], [51, 53, 55]]

In [62]: print(arr[0:6:2,1:6:2])

```
[[11 13 15]  
 [31 33 35]  
 [51 53 55]]
```

In [63]: [
 [40,41,42,43,44,45],
 [50,51,52,53,54,55]
]

Out[63]: [[40, 41, 42, 43, 44, 45], [50, 51, 52, 53, 54, 55]]

```
In [64]: print(arr[3:5:1,:])
```

```
[[40 41 42 43 44 45]
 [50 51 52 53 54 55]]
```

```
In [65]: [
    [14,15],
    [24,25],
    [34,35],
    [44,45],
    [54,55],
    [64,65]
]
```

```
Out[65]: [[14, 15], [24, 25], [34, 35], [44, 45], [54, 55], [64, 65]]
```

```
In [66]: print(arr[:,4:6:])
```

```
[[14 15]
 [24 25]
 [34 35]
 [44 45]
 [54 55]
 [64 65]]
```

```
In [ ]:
```